

Qatar-1b

R-2093

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121111 · created 2026-08-10T01:05:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456242.623348 +/- 9.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.036
Transit depth (Rp/Rs)^2	3.68 +/- 1.39 [%]
Orbital Inclination (inc)	84.68 +/- 2.13
Airmass coefficient 1 (a1)	0.764 +/- 0.069
Airmass coefficient 2 (a2)	0.198 +/- 0.058
Scatter in the residuals of the lightcurve fit is	9.17 %
Best Comparison Star	#2 - [528, 356]
Optimal Aperture	2.6
Optimal Annulus	5.82
Transit Duration (day)	0.074 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.036 vs 0.1463 (deviation 0.82 σ)
Duration fit vs published	0.074 d vs 0.0692 d (deviation 0.26 σ)
Mid-time O-C	0.1 min
Depth SNR	3.4
Residual scatter	9.17 %

Light curve

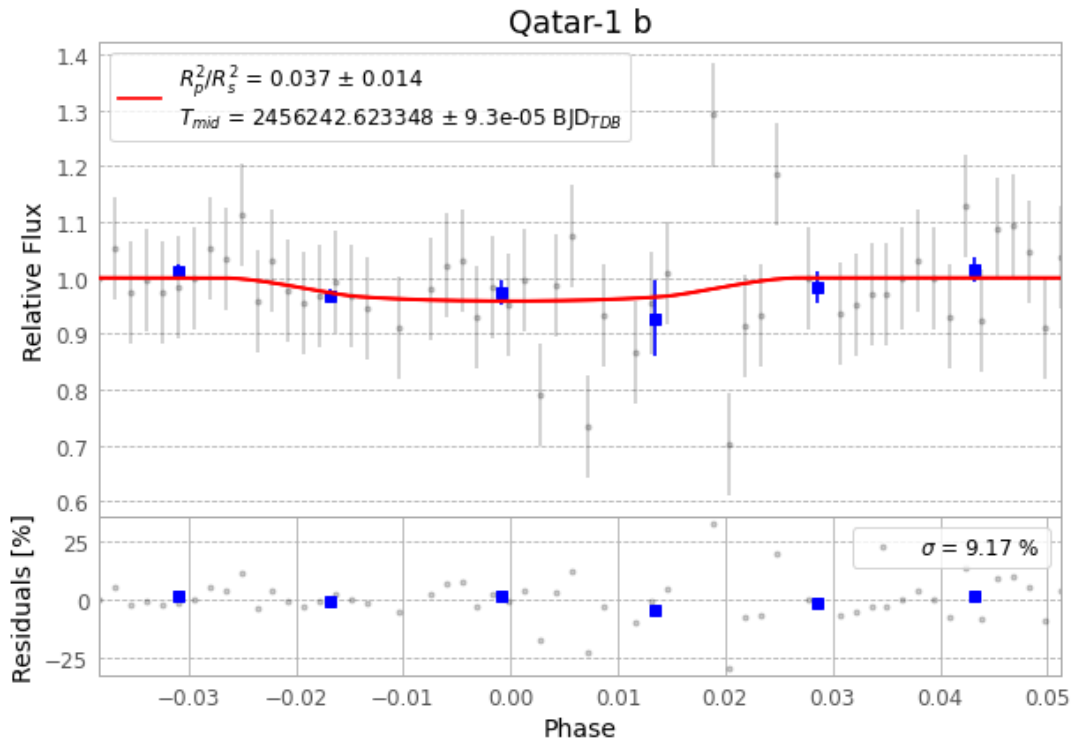


Figure 1 — FinalLightCurve_Qatar-1 b_2012-11-10.png (SHA-256 bf3fad44cf57c9dd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a06175331ac27683...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), Qatar-1121111013613.FITS → Qatar-1121111045112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121111.log (00720e5b00c8...), FinalLightCurve_Qatar-1 b_2012-11-10.png (bf3fad44cf57...), FinalLightCurve_Qatar-1 b_2012-11-10.csv (9fbfcf4728ca...)

Compute resources

Wall time 197.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 01:05Z.